# AtheroVision: Project Documentation

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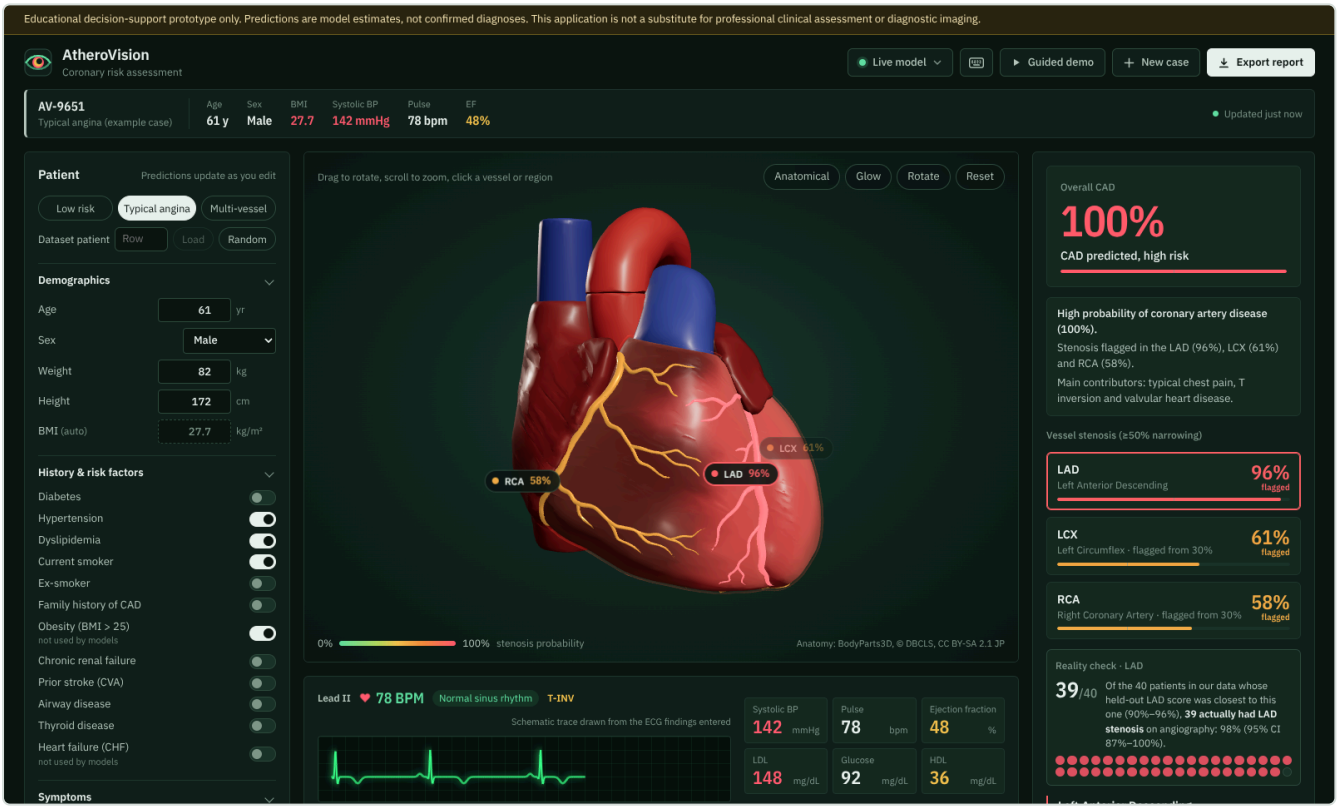
Multimodal AI Hackathon 2026, Track A: Cardiovascular Risk Visualization & Prediction.

## 1. Overview

AtheroVision predicts overall coronary artery disease (CAD) and ≥50% stenosis of the LAD, LCX and RCA from routine clinical data, and maps each prediction onto an interactive 3D heart. Four independent models trained on the 303-patient Z-Alizadeh Sani dataset reach a cross-validated ROC-AUC of 0.92 (CAD), 0.84 (LAD) and 0.74 (LCX, RCA).

AtheroVision integrates structured features from five clinical sources (demographics, clinical history and examination, ECG, laboratory results and echocardiography) and communicates its predictions through numerical probabilities, clinical explanations and interactive 3D anatomy. The CAD and LAD models use all five sources; the LCX and RCA models use seven history and demographic inputs. The models work on structured features, not raw ECG signals or images.

A clinician enters a patient (or loads an example or a real dataset patient). Within about 0.1 s the probabilities update and the three coronary arteries change colour on the anatomical model. Within about 2 s every score gets a SHAP explanation, a physiological breakdown and a *reality check*: how many similar-scoring patients in the data truly had the disease. A what-if panel, a printable report and a guided demo complete the workflow.

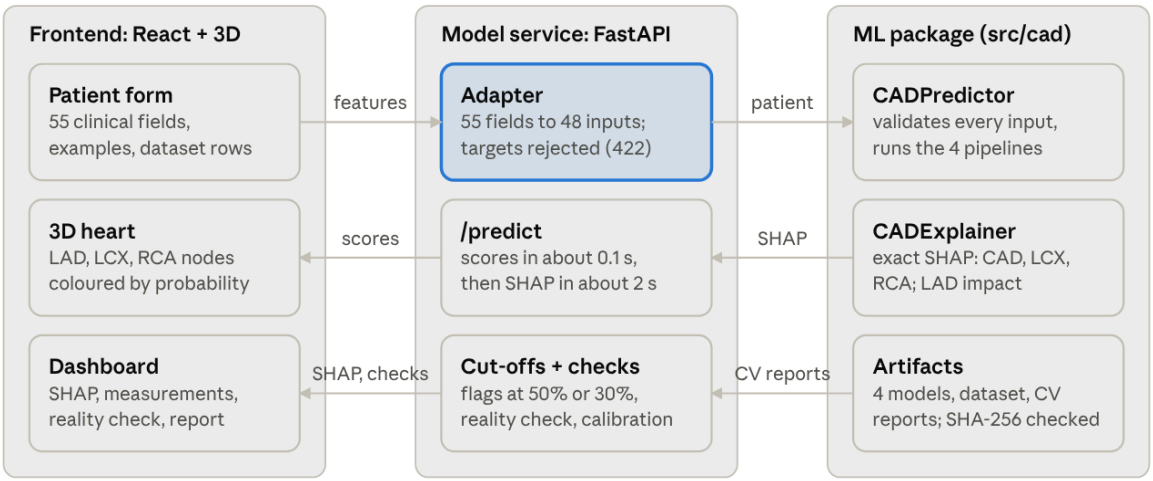



*The typical-angina example: LAD 96%, LCX 61%, RCA 58%, each artery coloured on the BodyParts3D heart. The visible disclaimer bar is always shown.*

## 2. System architecture

Three layers communicate over one HTTP request path. The browser never sees model files or secrets, and the service adds no modelling logic of its own: it calls the ML package's `CADPredictor` and `CADExplainer`.

**Features flow right, scores and explanations flow back to the heart**




The adapter (highlighted) is the leakage gate: it maps the form's 55 fields onto the models' 48 inputs and rejects any target column.

**Extending it without a redesign.**

- **A new clinical feature:** add one entry to the form schema (`features.ts`). Once a model is retrained with it, the adapter and input validation pick it up from the input schema in `config/model_selection.json`.
- **A better model for an existing target:** point the config at the new pipeline and its hash. Validation, explanations and metrics follow; a new target also needs an entry in the package's task list and a card in the dashboard.
- **A new anatomical structure:** add a named node to the GLB and map it to a target.

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## 3. Dataset and preprocessing

The models use the Extension of Z-Alizadeh Sani dataset (UCI): 303 patients, 55 inputs and four angiography labels, with no missing values. Prevalence is 71% for CAD, 58% for LAD, 39% for LCX and 38% for RCA.

- **Leakage control.** Cath, LAD, LCX and RCA are excluded from every model's inputs, as the brief requires. The API rejects them with HTTP 422 if they are sent, and tests check this.
- **Feature screening, decided before modelling.** Three columns were dropped as unlearnable: Exertional CP (constant), CHF (1 patient) and LowTH Ang (2 patients). That left 52 features (23 numeric, 29 categorical).
- **Redundancy.** BMI is computed from Weight and Height, Obesity is a BMI flag, and Neut mirrors Lymph ($\rho \approx -0.91$). The *reduced* set drops these four, leaving 48 features.
- **Preprocessing inside the model.** Each model is one scikit-learn Pipeline:
  - numeric columns: median imputation, then standard scaling;
  - categorical columns: most-frequent imputation, then one-hot encoding (one 0/1 column for two-level fields).

  The pipeline is refitted in every cross-validation fold, so test patients never influence preprocessing.
- **Data integrity.** The dataset file and every model file are checked against SHA-256 hashes in `config/model_selection.json` at start-up. A changed file stops the service.
- **Data quality.** Row 93 is labelled “no CAD” but has a stenotic LAD, which contradicts the dataset's own definition of CAD. It is kept as recorded (removing it changes CV AUC by at most 0.018), and the app flags it when that patient is loaded.

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## 4. Model architecture and validation

Each target has its own binary classifier, chosen from six candidates by a fixed simplicity rule: take the simplest model unless a more complex one is clearly better on the same folds.

| Target | Selected model | Inputs | Why |
|---|---|---|---|
| CAD | L2 logistic regression | 48 (reduced set) | Within noise of the forest (AUC 0.922 vs 0.927), better Brier score, exact explanations |
| LAD | Random forest (300 trees, min leaf 3) + sigmoid calibration | 48 (reduced set) | Highest AUC (0.853); beat logistic regression on most folds |
| LCX | L2 logistic regression | 7 clinical inputs | Highest AUC (0.743); the 52-feature model overfit (0.685) |
| RCA | L2 logistic regression | 7 clinical inputs | Highest AUC (0.740) and lowest Brier score |

**The six candidates, compared per target:**

- a no-skill baseline (always predicts the prevalence);
- logistic regression on 7 clinical inputs: age, sex, typical chest pain, diabetes, hypertension, dyslipidaemia, current smoking;
- logistic regression and random forest on all 52 features;
- logistic regression and random forest on the reduced 48.

No hyperparameters were tuned.

**Validation.**

- Repeated stratified 5-fold cross-validation, 5 repeats (25 train/test splits per target, seed 42).
- Every metric is computed on held-out folds only, with 95% bootstrap intervals (2,000 resamples).
- The LAD forest's probabilities were compressed (calibration slope 1.82), so a sigmoid calibration was fitted inside each training fold. That brought the slope to 1.00 with unchanged AUC.
- The deployed models were then refitted on all 303 patients.

**Flagging cut-offs.** Probabilities are the primary output. A target is *flagged* at the highest cut-off (0.05 steps, at most 0.5) that catches at least 80% of positive patients in cross-validation:

- CAD and LAD keep 0.5 (90% and 86% caught);
- LCX and RCA use 0.3, which raises the share caught from 52% to 84% (LCX) and from 43% to 83% (RCA), at the cost of more false alarms.

**Known optimism.** Models and cut-offs were chosen on the same folds they are reported on, so the estimates are slightly optimistic.

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## 5. Evaluation results

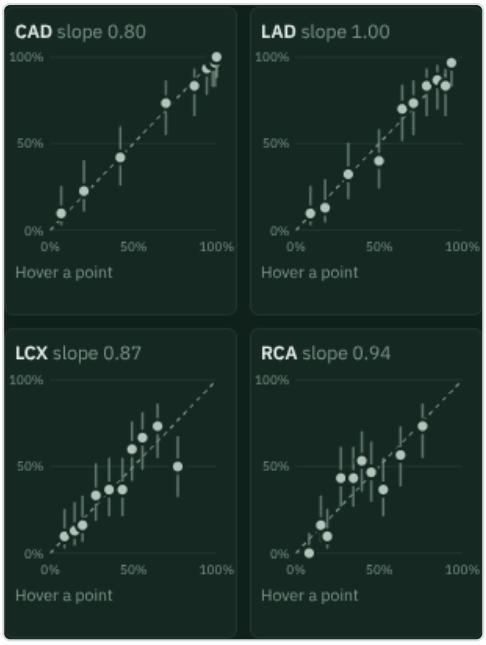
CAD is predicted well (ROC-AUC 0.92) and LAD reasonably (0.84); LCX and RCA are moderate (0.74) and need the screening cut-off. All values are cross-validated on 303 patients (5×5), with 95% bootstrap intervals for ROC-AUC. Accuracy, precision, recall and F1 use the 0.5 threshold; LAD is the deployed calibrated forest.

| Target | Prevalence | ROC-AUC (95% CI) | PR-AUC | Accuracy | Precision | Recall | F1 | Brier |
|---|---|---|---|---|---|---|---|---|
| CAD | 71% | **0.92** (0.89–0.95) | 0.97 | 0.86 | 0.90 | 0.90 | 0.90 | 0.10 |
| LAD | 58% | **0.84** (0.80–0.89) | — | 0.78 | 0.79 | 0.86 | 0.82 | 0.15 |
| LCX | 39% | **0.74** (0.68–0.80) | 0.60 | 0.69 | 0.62 | 0.52 | 0.57 | 0.20 |
| RCA | 38% | **0.74** (0.68–0.79) | 0.60 | 0.67 | 0.58 | 0.43 | 0.50 | 0.20 |

*PR-AUC was not recorded for the calibrated LAD model (0.88 for the uncalibrated forest; calibration does not change the ranking within a fold).*

At the screening cut-off of 0.3, recall rises to 0.84 (LCX) and 0.83 (RCA), and specificity falls to 0.54 and 0.53. Every model beats the no-skill baseline (ROC-AUC 0.50) by a wide margin.

**Risk estimation quality (calibration).** Calibration slopes are 0.80 (CAD), 1.00 (LAD), 0.87 (LCX) and 0.94 (RCA), where 1 is ideal; every 95% interval includes 1. So the predicted probabilities track observed rates and can be read as risk estimates within this population. CAD's slope below 1 means its extreme scores, such as 100%, are somewhat over-confident.




*Each point is about 30 patients grouped by cross-validated probability; the dashed line is perfect calibration. The same charts are in the app's Model performance panel.*

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## 6. Clinical interpretability

Every prediction is explained at three levels: per-input attributions, a physiological breakdown, and an observed-outcome check. The UI states on every panel that attributions are associations learned from 303 patients, not causes.

- **SHAP values (CAD, LCX, RCA).** For a logistic regression, the SHAP value of input *k* is $w_k \cdot (z_k - E[z_k])$ in log-odds, where z is the pipeline's own encoded input and E[z] the training average. AtheroVision computes this exactly, without sampling. One-hot columns are summed back to their clinical field. The values add up to the patient's score, and the service checks this to $10^{-8}$.
- **LAD (random forest).** Each input is replaced in turn by the values seen in the 303 training patients, and the average change in probability is reported. This is approximate and not additive; the UI labels it “not SHAP” and reports the remaining gap.
- **Physiological breakdown.** Each measurement is shown with its value, its normal range and status, and its share of the selected target's total attribution, as the brief requires.
- **Reality check.** For each score, the 40 dataset patients with the closest *cross-validated* score are found, and the app counts how many truly had the condition, with a Wilson 95% interval; a loaded dataset patient is left out of their own comparison. For example: “28 of 40 patients scored like this had LCX stenosis (70%, 55–82%).”
- **Held-out comparison.** Loading a real dataset patient shows the angiography result, the live score and the score the model gave while that patient was held out. Label conflicts (row 93) are flagged.
- **What-if simulator.** Modifiable factors (smoking, LDL, HDL, blood pressure, glucose, pulse, weight) can be changed; the four models re-run and the heart shows the scenario beside the current patient.

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## 7. 3D pipeline

The heart is a real anatomical mesh from BodyParts3D (DBCLS), converted once into a GLB file whose LAD, LCX and RCA nodes are exactly the three model targets.

1. **Build (offline, `tools/build_heart_model.py`).**
   - Loads the BodyParts3D heart wall, atria, great vessels and every coronary-artery segment.
   - Groups the artery segments by anatomical name into LAD (with septal branches), LCX and RCA, plus the left main trunk.
   - Re-orients and scales the result into the viewer's frame.
   - Stores territory weights in the ventricular wall's vertex colours, so each artery's supply region can be tinted.
   - Exports `heart.glb` (1.9 MB) with named nodes.

2. **Render (React Three Fiber / Three.js).** The viewer loads the GLB and finds the `LAD`, `LCX` and `RCA` nodes by name. Each artery's colour is set from its predicted probability on one shared scale: green at 0%, amber at 50%, red at 100%. The same function colours the dashboard bars, so the 3D view and the numbers always agree.
3. **Interaction.**
   - Drag to rotate and scroll to zoom.
   - Click an artery or its territory to select that target; the risk panel, SHAP chart and measurements follow the selection.
   - Labels show each artery's probability.
   - The heart beats at the patient's pulse rate.
   - The what-if scenario can be shown on the heart.
4. **No dedicated GPU required.**
   - The 3D code loads as its own chunk after the dashboard.
   - The renderer asks for the low-power GPU.
   - A performance monitor lowers the resolution and turns off the glow effect if the frame rate drops.
   - All automated browser tests ran on a software (CPU-only) renderer.
5. **Fallbacks and extension.**
   - Without the GLB, a procedural heart with the same three arteries is drawn.
   - A different mesh without named arteries can be aligned to the vessels in an Align panel.
   - Adding a structure means adding a named node and a mapping.

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## 8. Usage

On Windows with Python 3.12 or 3.13 and Node.js 20.19 or newer, double-click `start.bat`. The first run installs pinned packages; every run starts both servers and opens http://localhost:5173.

- **Manual start.** In `cardio_ai_hackathon`, run `python -m uvicorn api.main:app --port 8000`. In `frontend`, run `npm run dev`. API documentation is at http://localhost:8000/docs.
- **Workflow.**
  1. Enter or edit a patient, pick an example, or load a dataset row.
  2. Read the risk panel and the 3D heart.
  3. Inspect *Why this prediction*, the measurements and the reality check.
  4. Try the what-if simulator.
  5. Export the one-page report (E).
- **Guided demo.** Press P for an eight-step tour of every feature.
- **Demo mode.** `npm run dev:demo` runs the interface without the backend, and every result is labelled DEMO DATA.
- **Tests.** There are 94 automated tests. A GitHub Actions workflow runs them on every push:
  - `python -m unittest discover -s tests`: ML package, 39 tests;
  - `python -m unittest api.test_api`: model service, 27 tests;
  - `npm test`: frontend contract and logic, 28 tests.
- **Reproducibility.**
  - Exact package versions are in `requirements-backend.txt` and `package-lock.json`.
  - Fixed seeds (42).
  - Every model and the dataset are hash-checked at start-up.
  - All evaluation reports are in `cardio_ai_hackathon/reports/`.

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## 9. Limitations and clinical safety

AtheroVision is an educational decision-support prototype, and the interface says so in a disclaimer bar that is always visible: *"Predictions are model estimates, not confirmed diagnoses. This application is not a substitute for professional clinical assessment or diagnostic imaging."*

- **No external validation.** All results come from 303 patients at one centre; performance elsewhere is unknown.
- **Optimism.** Models and cut-offs were chosen on the same cross-validation folds they are reported on.

- **In-sample live scores.** The deployed models were refitted on all 303 patients, so live scores for dataset patients are optimistic. The app shows the held-out score beside them.
- **Moderate vessel models.** LCX and RCA reach a ROC-AUC of 0.74. Even at the 0.3 cut-off they miss about 1 in 6 stenotic arteries and raise many false alarms.
- **Independent models.** CAD and vessel predictions can disagree; the app shows the disagreement rather than forcing agreement.
- **Inputs from a cardiology work-up.** Echo ejection fraction and wall-motion abnormality are among the strongest inputs, so AtheroVision fits after a cardiology work-up, not first-contact screening.
- **Whole-vessel scores.** The dataset has no lesion locations: the 3D colour is a probability for the whole artery, never a lesion position or severity. RWMA is used only as an input, not as a location.
- **Associations, not causes.** Attributions and what-if results describe how the models respond, not the effect of a treatment.
- **Privacy.** Everything runs locally; no patient data leaves the machine.

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## 10. Sources and credits

- Dataset: Extension of Z-Alizadeh Sani Dataset, UCI Machine Learning Repository (303 patients; adds the LAD, LCX and RCA labels).
- 3D anatomy: BodyParts3D, © The Database Center for Life Science (DBCLS), CC BY-SA 2.1 Japan. `heart.glb` is a derivative under the same licence.
- Software: scikit-learn, pandas, FastAPI, React, Three.js / React Three Fiber, Vite.
- Detailed evaluation: `MODEL_CARD.md` and `cardio_ai_hackathon/reports/` in the repository.